Effect of Nano-Formulated Agrochemicals on Rhizospheric Communities in Millets

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Abstract

Nanotechnology is an emerging technology that combines materials science and engineering fundamentals, altogether synthesizing materials at the nanoscale to solve various current problems. Agrochemicals are often used in modern intensive agriculture to protect crops from biotic stress, provide vital nutrients, and boost millet growth and yield. Although they are beneficial in the short term, their long-term and persistent applications damage soil fertility and negatively affect the rhizospheric microbiome. Nanotechnology in the form of nano-based agro-formulations is an innovative, environmentally acceptable, and practical solution for substitute synthetic fertilizers. Nano-technological formulations and nano-particles in the form of nano-pesticides, nano-herbicides, nano-gels, nano fertilizer, and nano-fungicides are reported to have efficacy in continuous release of nutrients regulated distribution to plant nutrients very effectively that helps in fighting phytopathological diseases, promotes plant and microbiome productivity. In addition, they are effective in alleviating biotic and abiotic stress in crop plants. However, despite their extraordinary effectiveness, they also have several drawbacks, such as a tedious manufacturing process, shaky delivery, and dosage-sensitive effectiveness. Learning and acknowledging the influence of nano-based agrochemicals on a highly nutritious crops like millets is important for further commercialization and wide utilization in agriculture. Hence, this chapter focuses on the usage of nanoformulations on millet crops, their effects on soil rhizospheric communities, and soil fertility.

Keywords: Nanoformulations, rhizosphere, millets, soil biota, millet-grown soils

1. Introduction

The sustainable increase in food production and eradication of hunger and poverty are the main goals of agricultural sustainability and food safety in world countries. The inorganic synthetic fertilizers provide macronutrients and micronutrients for plant growth, but many agrochemical applications in agricultural fields have shown several disadvantages in the long term. Nutrient Use Efficiency (NUE) of soil reduces with continuous crop growth and application of various chemicals in the soil, leading to imbalanced nutrients and low nutrient uptake by plants. More than 90% of the applied chemicals are lost in the environment, while others are unable to reach the target locations required for efficient use (Nizzetto et al. 2016). For many years, fertilizers and insecticides have been crucial in ensuring food production demands are met (Guo et al. 2018). However, excessive usage of these chemicals also impacts sustainable agriculture development. Nanotechnology has revolutionized its application in agriculture, medicine, and pharmaceuticals (Raliya et al. 2017; Singh et al. 2021). Nano-based agrochemical formulations are introduced in the agroindustry to reduce the excessive utilization of chemical fertilizers and improve soil nutrition with minor alternations to soil properties and microbial communities (Arora et al. 2022). The engineered nanoparticles such as nano fertilizers, nano pesticides, nano insecticides, nano rodenticides, and other nano formulations are utilized in agriculture for their broader scope, and distinct advantages such as controlled release of loaded substances, loss of nutrients, bioavailability, solubility, and desirable synthesis methods and are found as alternatives for conventional chemical fertilizers. A very limited number of highly tiny particles are used in the nano-formulation of agricultural chemicals, which act on the plant rhizosphere as active ingredients in fertilizers or other agricultural chemicals (Ul Haq and Ijaz 2019). They have been extensively tested for their positive and negative impacts on soil health for the past two decades since they are considered as future enhancers of food production (Kumari and Singh 2020; An et al. 2022).

Millets are the most popular crops grown in arid and semi-arid regions of Asia and southern Europe. The agrochemicals such as fertilizers and pesticides have been reported to influence the yield, soil fertility, microbial communities, and enzymatic activities in agriculture. The rhizosphere communities of millets are less focused than rice, wheat, and maize, and these microbial populations participate in all biogeochemical cycles and enhance the nutrient availability to millets (Xu et al. 2018). The first part of this chapter focuses on the influence of agrochemicals and nanoformulations on soil microbiota (rhizosphere communities, soil properties, soil fauna). The latter focuses on existing nano-based agro-formulations and possible impacts on millet rhizosphere communities.

2. Current trends in Nano-agrochemical formulations

In recent years, nanotechnology interventions in agriculture have extended substantially, including nanoscale carriers, nano sensors, clay nanoparticles, nano polymeric composites, nano emulsions, and nano food packaging materials. These novel technologies are also utilized to combat plant diseases, identify pathogens, boost plant resistance, and reduce post-harvest losses. Nano-based agrochemical formulations provide a wide range of benefits (more effectiveness and durability, better distribution and permeability, ability to disintegrate in the soil and environment, lack of toxicity, and photo generative nature) as well as a reduced amount of agro-inputs with convenient pesticide properties to protect crops against insect pests and diseases effectively **(Fig. 1).**

Nano carriers are utilized for slow and sustained delivery of fertilizers, fungicides, herbicides, pesticides, growth regulators, and other target active compounds (Vega-Vásquez et al. 2020). Nano polymeric compounds from plants, animals, and microbe-based by-products create opportunities for exploring innovative and value-added nano materials and products. For example, clay-based nanomaterials can serve as versatile carriers of nutrients and pesticides with the advantage of extended release and could reduce 70-80% of pesticide use (Merino et al. 2020).

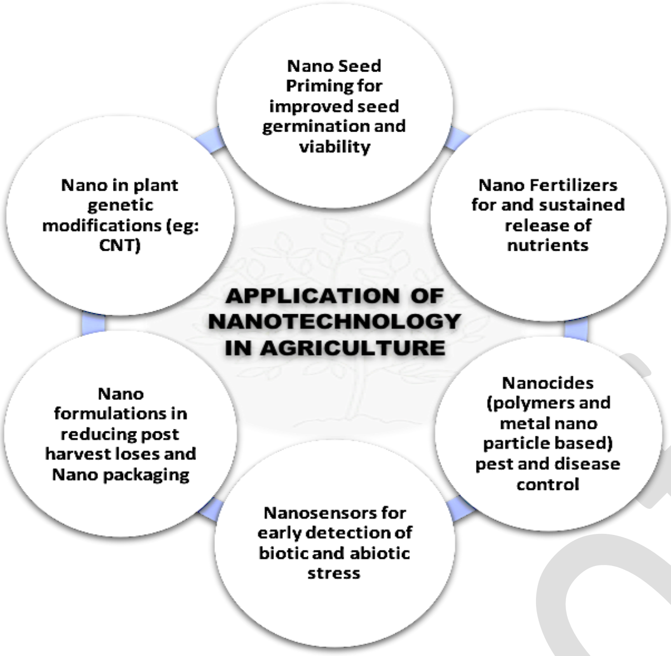

Fig. 1. Application of nanotechnology in agriculture

2.1. Nano carriers for Agrochemical formulations

Potential applications of nanotechnology for integrated pest management include the controlled administration of Artificial Intelligence (AIs) with increased activity at lessening drug concentrations. Nano carriers significantly improve the chemical stability of agrochemicals while protecting them from harsh environmental factors (such as radiation and high temperatures) (Kumar et al. 2019). Nano encapsulation, also known as an encapsulation of active ingredients with nanoparticles, is a technique for the sustained but effective release of chemicals to a specific host plant. The biphasic dispersion of two or more immiscible liquids (water in oil or oil in water) to form droplets stabilized using any amphiphilic surfactant to create an emulsion droplet less than 100 nm is referred to as nanoemulsions. Nanoemulsions are transparent or translucent oil-water emulsions used in agriculture to promote physio-chemical stability, improve the properties of pesticides (bioavailability), and sustainable release of chemicals (Bhattacharyya et al. 2016).

The chemical or physical confinement of gels in nano size, which serves as nanocarriers for delivery of agrochemicals or drugs, is referred to as nanogels. These nanogels are swellable cross-linked polymer networks with a high ability to hold water without dissolving it into an aqueous medium. These polymers are either natural or synthetic and combination of both with properties such as amphiphilicity, softness, porosity, and degradability according to the variability in composition. The materials (i.e., nanoparticles) loaded into the nanogels protect the loaded material from degradation and elimination, and additionally, they actively participate in the delivery at the target site. The stimuli-responsive behavior of nanogels with stability makes them suitable for the carrier of desired nanoparticles. Polymeric nanogels are a new hybrid to overcome poor colloidal stability, elimination, and low aqueous solubility. Aide variety of molecules can be incorporated to increase their utility (Soni et al. 2016). For instance, the core-shell type lignin nanogels formed using lignin methacrylate, emulsified and graft copolymerized along with acrylic acid, were stimuli-responsive (pH-responsive swelling). Therefore, lignin bio-based nanogels could be used as a nanocarrier for the controlled release of pesticides or fungicides (Yiamsawas et al. 2021). The different types of nano- agrochemical formulations are used in the manufacturing of nano-fertilizer, nano-pesticides, nano-herbicides, nano-fungicides, etc. **(Fig. 2)**

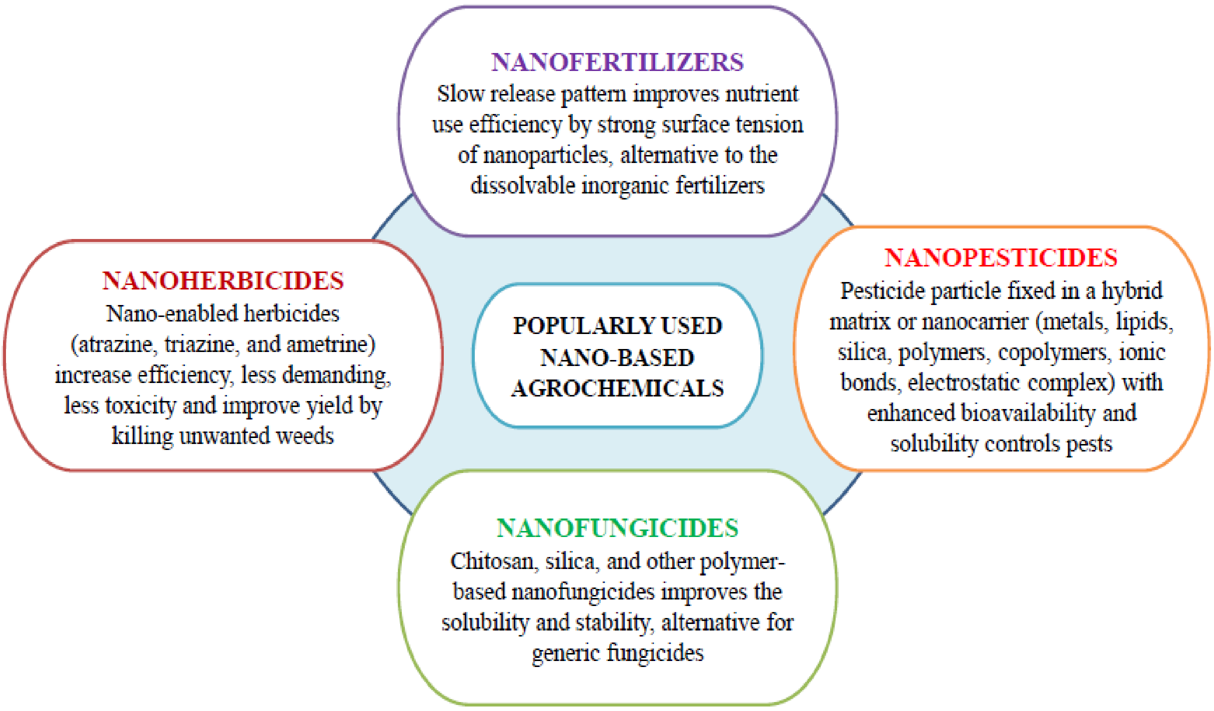

Fig. 3. Commercially used nano-based agrochemicals

3. Soil rhizosphere communities: Role, interactions, and importance in crop production

Nanotechnology provides a novel agrochemical tool that helps in high crop productivity, hazardous chemical reduction, plant-pathogen management, and crop protection through plant adaptation (Shang et al. 2019). Nanoparticles may have an impact on the soil microorganisms in the environment via (1) physical direct contact, (2) alteration of available nutrients and toxins, and (3) indirect effects due to their interaction with soil organic materials, therefore, leading to alterations in cell surface morphology and growth behavior, disruption of biofilm and core microorganisms of the millet rhizosphere including many organisms (Xie et al. 2017). Microbial communities in the rhizosphere are well known to be closely linked to plant roots. In addition, rhizosphere bacteria significantly impact nutrient uptake by increasing or lowering nutrient availability. Microbial communities in the rhizosphere are a subset of the soil microbial community but are frequently distinct from those in the bulk soil (Dotaniya and Meena 2015).

Furthermore, rhizosphere microorganisms are recognized to have a crucial role in plant health, and plant health influences rhizosphere microbes in millets (Babu et al. 2019). It has been found that the rhizosphere microbial communities associated with healthy and diseased plants differ, although rhizospheres associated with infected and dead plants have comparable microbial compositions (Liu et al. 2020). In millets, the rhizosphere is the most active micro-domain for plant-microbe interactions, and it contains microorganisms vital for plant growth and health maintenance. Rhizosphere bacteria, for example, can provide nutrients to plants by speeding up nutrient cycling or nitrogen fixation and generating hormones or antibiotics that improve plant drought, salt, and disease tolerance.

The microbiota can either hurt or help the host plant. Beneficial microbes in the soil rhizosphere can aid plant adaptation to be natural and artificial environmental challenges and climate change (Zhou et al. 2009). Microorganisms in the rhizosphere have positive and negative interactions with plants and roots because soil habitats vary due to various circumstances, defining the microbiome for millets becomes critical (Elhady et al. 2018). Due to their quick sensitivity to environmental changes, microorganisms serve as early warning biological indices of soil and root health. Microbial diversity in the rhizosphere fluctuates as a result of such variables, as well as farming patterns and climate (Rosen and Allan 2007). The microbial diversity in the rhizosphere of finger millet is unknown at this time; thus, the millet microbiome is essential for future nutritional and food security.

Furthermore, the finger millet plant indicates its potential involvement in reducing pathogenic strains and offering medicinal and nutritional benefits to the linked plants. In environmentally problematic settings such as high salinity and locations with severe water shortages, it would be worthwhile to include beneficial species of this genus as bioinoculants to boost stress tolerance and produce finger millet (Kour et al. 2020). Intraspecific variability of millet rhizodeposition and, more broadly, phenomena leading to root-driven soil aggregation and microbiological diversity are significant guides to long-term studies on climatic stress and early warning cues of aphids and other pathogen attacks. To measure the diversity of the ratio between the mass of root-adhering soil and the mass of root tissue, researchers looked at the relationships between soil adherence to roots, aggregation, microbiological activity, and root exudation (Amellal et al. 1998). Furthermore, the nutritional quality of the sediment and the vegetation mix of the patches may influence plant species' ability to shape their rhizosphere microbial community (Zhang et al. 2020). Microbes in the rhizosphere are essential for plant health and ecosystem functioning in terrestrial ecosystems. Yet, our knowledge of microbial co-occurrences and assembly processes in the millet rhizosphere is limited.

4. Influence of agrochemicals on soil microbiota

The unsustainable agricultural practices in many developing countries resulted in direct and indirect ecosystem contamination. The new technologies have further influenced the adverse impact of unsustainable practices, leading to land, water, and atmospheric degradation. The various agrochemicals such as herbicides, pesticides, insecticides, molluscicides, nematicides, rodenticides, fungicides, and many other chemical or inorganic compounds have impacted soil microbiota (Meena et al. 2020). A large quantity of agrochemicals used in agriculture has affected soil microbiota directly, which are considered a biological indicator for soil fertility. The nutrient status of soil, microbial carbon biomass, and non-target biota are disturbed by the impact of various pesticides. They also affect the soil enzymes such as urease, dehydrogenase, nitrate reductase, oxidoreductase, and hydrolyses (Wang et al. 2006; Hussain et al. 2009).

4.1 Effects of fungicides on soil microbiota

The harmful effects on microbial growth, functions, metabolism, and survival alter soil fertility. Some of the negative impacts of fungicides on microbiota are: (i). mancozeb and chlorothalonil could reduce the nitrification and denitrification process; (ii). Carbendazim is highly toxic to a potential biocontrol, *Trichoderma harzianum* (anti-fungal against Fusarium, Rhizoctonia, and Pythium); (iii). Copper-based fungicides affect the N-fixing bacterial population; (iv). Dimethomorph inhibits ammonification and nitrification in sandy soil; (v). Fludioxonil is toxic to algae; (vi). Captan decreases nitrogenase activity and reduces viable counts of nitrifying and denitrifying bacteria, *Rhizobium cicero; (*vi). Dinocap affects the ammonifying bacteria (Kyei-Boahen et al. 2001; Virág et al. 2007; Cycoń et al. 2010). In addition, my non-target fungicides cause inhibitory effects by reducing the beneficial microbiota in soil .

4.2. Impact of herbicides on soil microbiota

Based upon the type of molecule present in the applied herbicide, the microbial population decreases within 7 to 30 days. Microbes' biosynthetic mechanisms or physiology are affected indirectly, affecting soil enzymes, protein biosynthesis, and plant growth regulators (Ethylene, gibberellins, and Indoleacetic Acid) (Kremer and Means 2009; Meena et al. 2020). The effects of herbicides on soil microbiota depend on concentration, degradability, adsorption, desorption, persistence, bioactivity, and toxic dosage level. Some microbes which can withstand a single application (Butachlor) are more sensitive to combination (Butachlor with cadmium) (Hussain et al. 2009). The inorganic fertilizers combined with heavy metals suppress soil microbial activities, and secondary metabolites produced by the transformation of herbicides are more severe than direct effects. For instance, metabolites produced by 2,4-D affected a group of gram-negative bacteria, *Burkholderia cepacia.* Even after the N fertilization, prosulfuron in soil affects the nitrification and denitrification process. Atrazine and simazine (10 mg/L) were toxic for the growth and biological activities of Xanthobacter autotrophic. Acetochlor and its derivatives negatively impact bacterial communities such as *florescens*, *Mycobacterium phlei, and Bacillus subtilis*. The enzymatic activities such as dehydrogenase and invertase are affected by atrazine and metolachlor (Sáez et al. 2006).

4.3. Effects of insecticides on soil microbiota

The growth and population of bacterial species such as Azotobacter, Pseudomonas, Bacillus, Penicillium, and many others are affected by long-term insecticides such as fenthion parathion and malathion. The soil polluted with 15 organochlorine pesticides exerted severe ecological risk, among which 76% of ecotoxicity was exhibited by endosulfan. The bioindicators, including 61005 bacterial species and 33397 fungal species were not found in organochlorine pesticide-affected soil, whereas they were present in uncontaminated soil (Egbe et al. 2021). In addition, the short-term application of thiamethoxam reduced the microbial diversity, bacterial abundance, and structure during short-term application (Actinobacteria) (Wu et al. 2021). Like other pesticides, insecticides inhibit the soil enzyme activities (urease, phosphatase, and nitrogenase), microbial biomass, and beneficial species population reduction.

5. Application of Nano-formulated agrochemicals for various crops

5.1. Enhancing crop quality and growth using nano-formulated fertilizers

Nano-fertilizers are a noteworthy recent development in decreasing the loss of applied fertilizers to the environment. By combining plant nutrients with nanomaterials, coating nutrient molecules with a thin layer of nanomaterials, and creating nano-sized emulsions, nano-fertilizers are created that intelligently increase soil fertility and bioavailability (Sidorowicz et al. 2019). Additionally, studies of nutrients on slow/controlled release or control loss of nano-fertilizers conducted in soil and water have confirmed that the long-term availability of all doped nutrients to the plant over the whole crop period of cultivation is essential for promoting germination, growth, flowering, and fruiting (Lateef et al. 2016). Different nanomaterials have been used to encapsulate macronutrients like carbon (C), nitrogen (N), potassium (K), phosphorus (P), calcium (Ca), sulphur (S), and magnesium (Mg) to increase crop fertilizer uptake and reduce fertilizer outflow (Singh et al. 2021). For instance, urea-hydroxyapatite (HA) Nano formulations have the potential to increase release time and decrease nitrogen fertilizer consumption. Recent research has demonstrated that NPs, including CNTs, silicon dioxide (SiO_2) NPs, zinc oxide (ZnO) NPs, titanium dioxide (TiO_2) NPs, and even gold (Au) NPs, have favorable impacts on seed germination in agricultural plants, including tomato, wheat, rice, pearl millet, soybean, barley, and maize (Singh et al. 2016). Compared to untreated plants, the germination rate of pearl millet was dramatically improved by Au Nano formulations (Parveen et al. 2016). Numerous studies have demonstrated the benefits of using nanoparticles on germination, plant growth, and development (**Table 1**).

Manufacturing proteins, carbohydrates, and hormones (such as auxins) are all regulated by micronutrients, which shield plants from infections and pests. Fe/SiO_2 nanoparticles have a great potential to enhance seed germination in barley and maize (Najafi Disfani et al. 2017). In addition to germination, nanomaterials such as ZnO, TiO_2, MWCNTs, FeO, $CuZnFe_4O_4$-oxide, and hydroxyl fullerenes are said to promote crop growth and development with quality improvement in a variety of crop species, including mustard, onion, spinach, tomato, potato, and wheat (Shalaby et al. 2016). For example, Zn nanoparticles were used to boost grain output in a field crop called pearl millet (*Pennisetum americanum* L.) by roughly 38% (Moghaddasi et al. 2017). In several horticultural crop plants, including garden pea, cucumber, spinach, tomato, potato, eggplants, chilli, coriander, and onion, the usefulness of zinc-based nano fertilizers is documented (Ahmed et al. 2018; Pullagurala et al. 2018) (**Table 1**).

5.2. Nano-formulated chemicals for crop development and protection

Insecticide, herbicide, fungicide, and bactericide nano-formulation or encapsulation using nanomaterials holds immense potential for reducing agrochemical doses, boosting crop productivity, and fostering sustainable development (Liu and Lal 2015). Various nano-formulations of agricultural chemicals are summarized in **Table 2**. Additionally, using fewer pesticides lowers the cost of growing crops. In these situations, the application of NPs is reported as an effective alternative to directly decrease pest and disease infection and activity, resulting in higher crop development and output (Servin et al. 2015). Polymeric NPs (like chitosan and solid lipids), inorganic nonmetallic NPs (like silica NPs and nanoclays), and metallic NPs (like Cu NPs and ZnO NPs) are examples of agrochemical nanocarriers (Shukla et al. 2020).

Furthermore, numerous studies have demonstrated that nano-formulated pesticides are more effective at eliminating pests without harming people or the environment. For instance, when applied to *Drosophila melanogaster*, spinosad- and permethrin-loaded chitosan NPs showed enhanced bioavailability even at lower dosages than free spinosad and free permethrin, and the nanocomposites caused less harm to people and the environment (Sharma et al. 2019). The nano-insecticide particles shrink and become more concentrated after being encapsulated by NPs, giving them stability and gradual release capability. Furthermore, pesticides can exert systemic effects to suppress chewing or sucking-type insects like aphids by increasing their ability to penetrate through plants and reach the cell sap (Li et al. 2007). For instance, several nano-insecticides benefit from the poisonous properties of metallic NPs. Compared to bulk Al_2O_3 treatment, aluminum oxide (Al_2O_3) NPs showed excellent capability for eradicating *Sitophilus oryzae* on stored rice (Das et al. 2019). It has been established that using pheromones to manage insect populations is a promising and successful strategy (Dweck et al. 2015).

The foliar application of polysaccharide NPs loaded with metsulfuron-methyl to wheat-growing weeds dramatically reduced weed biomass compared to conventional pesticides. Additionally, the cytotoxicity of conventional and nano-herbicides was determined by culturing cells, and the results revealed that the herbicide-loaded NPs were less hazardous than the conventional herbicides (Kumar et al. 2017). According to a different study, solid lipid NP-based nanoherbicides had improved release patterns and herbicidal action than conventional herbicides. *Raphanus raphanistrum* (a weed species) development was considerably slowed down when atrazine was encapsulated by solid lipid NPs as opposed to the untreated group.

Additionally, *Zea mays* were unaffected by the tested nanoherbicide concentration (de Oliveira et al. 2015). In weeds treated to SiO_2 nanoparticles, (Sharifi-Rad et al. 2018) found that germination, root and shoot lengths, fresh and dried weights, and photosynthetic pigments with total protein all fell dramatically. Similar to this, (Kumar et al. 2017) demonstrated that pectin (polysaccharide) nanoparticles loaded with the herbicide (metsulfuron methyl) are more cytotoxic to *Chenopodium album* plants both in the laboratory and in field conditions, and only a very low amount of AI is needed in comparison to the commercial herbicide. Commercial herbicides only control or eradicate the weeds' above-ground sections, leaving their underground components like rhizomes or tubers unaffected. As a result, weeds begin to grow again; however, nanoherbicides stop this from happening (Dwivedi et al. 2016).

6. Impact of nanoformulations on plant growth (Root and plant endophytes)

The interaction and uptake of nanoparticles are influenced by several factors like the nature of nanoparticles, plant physiology, the interaction of nanomaterials with the environment, etc. The mechanism of entry of nanoparticles, their mode of application, and their penetration through plant cells are detailed in **Fig. 3.** The common factors influencing nanoparticle uptake and transport in plants are, a) Nature of nanoparticle, its surface charge, and chemical properties; b) Method of application of nanoparticle; c) Nanoparticles interact with microorganism in soil, which might impede their absorption into the plant system via several tissues like epidermis, endodermis, casparian strips, and cuticle till vascular bundles; d) Mechanisms involved in nanoparticle internalization within the plant tissues, such as mediation of carrier proteins, pore formation, endocytosis, etc. (Pérez-de-Luque 2017).

The entry of Nanoparticles occurs via. root system through cells using symplastic pathway or around the cells to reach xylem via apoplastic pathway (Lv et al. 2019). The nanoparticle uptake involves diffusion, gated channel confirmation, symporter, and antiporter active transport. Diffusion involves the movement of ion molecules from a region of higher concentration to a region of lower concentration. There are two types of gated ion channels, one, the Voltage-gated ion channels, and two, ligand-gated ion channels. These are the class of transmembrane proteins. In the case of voltage-gated ion channels, it forms the ion channel activated near the channel by changes in the electrical membrane potential, which alters the conformation of the channel proteins, regulating their opening and closing. While in the case of ligand-gated ion channels, they provide a means of converting the biochemical signals into ion flux and electrical events near the membrane (Banerjee et al. 2019). In plants, the intracellular and extracellular signaling ligands are represented by cyclic nucleotides and glutamate, respectively.

A symporter and the antiporters are integral membrane proteins and a type of co-transporter and exchanger, respectively. The symporter in the plasma membrane involves in exchange of two dissimilar molecules in the same direction through the cell membrane, whereas the antiporter exchange two or more different molecules or ions across a phospholipid membrane termed "secondary active transport" in opposite directions, one into the cell and one out of the cell (Lv et al. 2019).

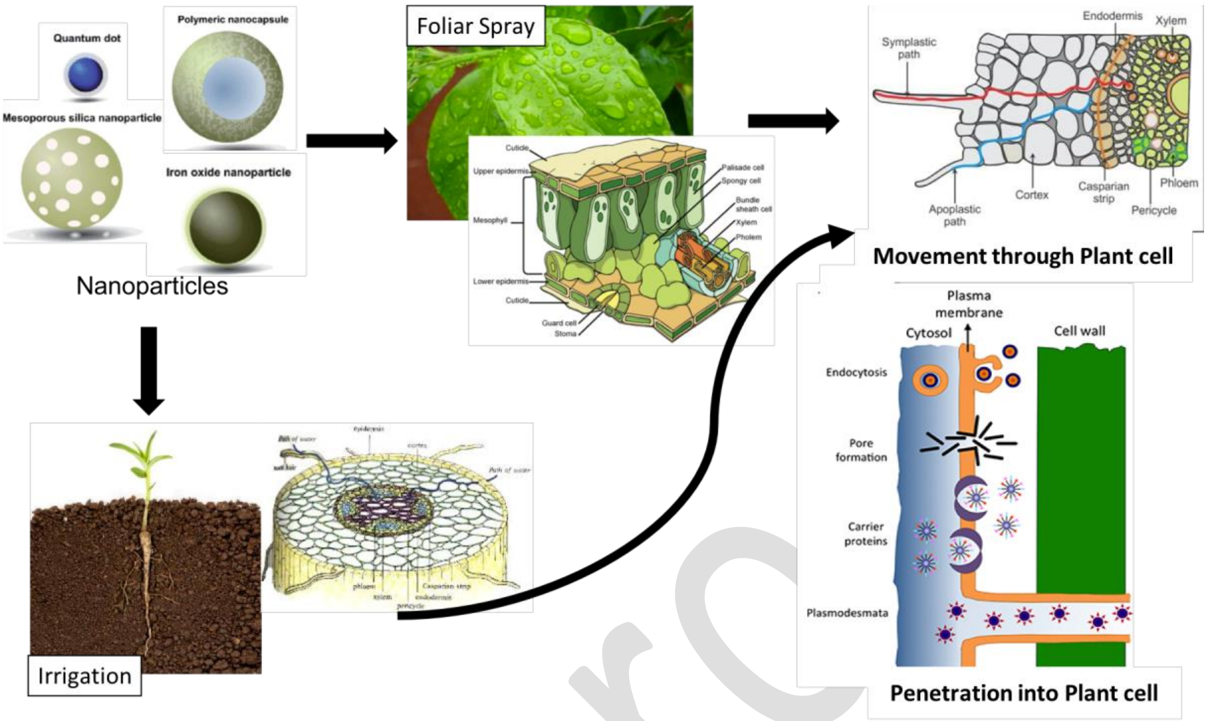

Fig.3. Interaction of nanoparticles in soil-plant interface

7. Impact of nano-formulated agrochemicals on soil health

7.1. Soil properties

Soil structure and functions are vital systems for ecosystem sustainability. The primary sink for all the chemicals applied for crop growth and production is soil and sediments. Clay composition, pH, temperature, minerals, and organic matter play an adverse role in agrochemical absorption and activities in soil. The interaction between nanoformulations and soil decides the leaching, runoff, bioavailability, distribution, uptake, and action. Or else, chemical residues remain in the soil for years inhibiting microbial and other enzymatic activities in soil due to long-term persistence. For instance, alachlor and metolachlor were reported to persist in soil water and were detected after 39 and 18 months, respectively (Vryzas 2018). A slow-release nano fertilizer- Chitosan nanoparticles incorporated with potassium improved the soil's physical properties such as high water conductivity, porosity, and enhanced friability, supporting Zea mays' root growth. *I*ts dry density reduction induced the higher uptake of nutrients by plants and doubled the root biomass. Better carbon-cycling activities, higher microbial activities, and soil conditioning by stabilizing the soil aggregates and cohesive bonding of soil particles prevented soil degradation. As a result, ty increased productivity with sustained release of nutrients (Kubavat et al. 2020).

Although nano-based agrochemicals tend to have many advantages on soil properties, it also exhibits adverse effects. For example, nano fertilizer was observed to decrease the soil pH level and increase exchangeable potassium but has no significant impact on total nitrogen and available phosphorous (Pide et al. 2022). On the other hand, nanocarriers used to deliver fungicides, insecticides, and fertilizers anchor the plant roots to soil organic matter and structure to reduce the runoff of applied chemicals (Bratovcic et al. 2021). Therefore, a sustainable and slow-release pattern of nano-formulations can be achieved based on the soil structure and properties.

7.2. Rhizospheric interaction and microbial communities

Agrochemical formulated nanoparticles may have a noticeable impact on the rhizospheric communities of soil. It has been discovered that the uptake of synthetic agrochemical nanoparticles into roots and root nodules eliminates N2 fixation potentials and impairs the growth of numerous crops (Priester et al. 2012). Investigations have shown that carbon-based and chemical NPs can produce stress, generating excess ROS that can affect many plants' proteins, lipids, carbohydrates, and DNA (Zhai 2020). Carbon nanotubes, one of the carbon-based ENPs, have been reported to induce ROS accumulation, enhancing lipid peroxidation in cell culture (Liu et al. 2010) and seedlings' root tips. Agrochemical formulations' effects on Rhizobium in crops were also studied by (Fan and Smith 2021), who discovered that nano agrochemicals cause morphological alterations in bacterial cells. Chlorophyll a, the major photosynthetic pigment in plants, is more sensitive to photodegradation than other pigments and could be a more useful indicator of agrochemical NPs toxicity compared with growth characteristics of the crops and rhizospheric communities.

Furthermore, the growth of root nodules and the consequent delay in initiating nitrogen fixation were observed as signs that the relationship between these microbial organisms was disrupted. According to reports, modifying bacterial communities was dose-dependent, with some taxa increasing as a percentage of the community while more taxa decreased, leading to lower diversity. The soil bacteria may be in contact with NPs when applied directly to land containing mobile NPs. Some of these microorganisms are also effective at adsorbing and accumulating nanomaterials of one kind or another, which starts the mobilization of nanomaterials via food chains and can change communities of different populations (Holden et al. 2013). The well-known agricultural microbicide NPs has a detrimental effect on plant growth and destroys the soil microorganisms supporting it (Yadav et al. 2018). Nano formulations were found to drastically inhibit the activity of various soil enzymes, including soil protease, catalase, and peroxidase, in addition to microorganisms (Du et al. 2014). Additionally, it was shown that inorganic nanoparticles harmed soil microbial communities. These microbes' toxicity is further significantly enhanced by the presence of light (Adams et al. 2020).

The unique toxicity can be attributable to the nanoparticles and their microbial interactions, which correlate the physicochemical features of agrochemical formulations with their biological response species. However, the material's surface coating, which environmental factors can considerably change, can reduce or increase microbial toxicity (Suresh et al. 2013). NPs can be ingested by microorganisms, according to studies on ecologically significant bacterial species like *E. coli*, *Bacillus subtilis*, *Pseudomonas putida*, and others. Most microorganisms had learned how to efflux, detoxify, and accumulate the metal ions long before the plants. They also created efficient molecular mechanisms and operated specific metabolic pathways. Additionally, bacteria can volatilize certain ions to remove their acute toxicity (De Souza et al. 1999). More research is required in agrochemical formulations for millets regarding beneficial soil microbes like N_2 fixing, phosphate solubilizers, and AM fungi to establish the uptake mechanisms and outcomes in soil and microbe interactions, even though microbes have evolved resistance and avoidance mechanisms.

7.3. Soil fauna

Nanoformulations, including nano zero-valent iron (nZVI), have been utilized for years to remediate polluted soils and aquifers. However, little research has examined its secondary environmental consequences and ecotoxicity on soil organisms (El-Temsah and Joner 2012). Hence, the translocation of NMs into microorganisms, soil fauna, edible food tissues and their influence on food quality and safety are critical concerns in nanotoxicology research (Zhao et al. 2014). Soil fauna may also change rhizosphere soil's physical and biochemical environment (Bray et al. 2019). Hence, soil fauna is regarded as an ecosystem engineer (Cunha et al. 2016) and plays a significant role in soil processes, including soil formation, waste decomposition, and nitrogen mineralization (Bardgett and Chan 1999). Large numbers of microorganisms (e.g., bacteria and fungus) and soil fauna (e.g., earthworms) assemble in the rhizosphere around plant roots, producing a distinct biotic and abiotic environment via root secretion, microbial growth, and earthworm activity in soils (Berendsen et al. 2012). Earthworms are ubiquitous in many types of soil and makeup 60–80% of the total soil biomass, making them excellent for soil toxicity studies (Bouché 1992). Earthworms are delicate, simple to handle, and toxicity test data is accessible. *E. fetida* is an easy-to-cultivate test species (OECD 1986). However, earthworm research on the transport and availability of NMs in soils is sparse. X-ray tomography demonstrated that earthworm burrowing governed nanoparticle movement and fate in soil (Baccaro et al., 2019). Numerous studies show that earthworms promote the abundance and activity of soil microbes (e.g., PGPR, plant symbiotic microbes) in the rhizosphere by burrowing and secreting mucus (Braga et al. 2016). This indirectly mediates the geochemical behaviour of NMs driven by microbes in the rhizosphere.

NMs interact with soil components (minerals and organic matter) before entering plants; hence their bioavailability may be affected by soil physicochemical parameters, especially pH and ionic strength of soil pore water (Tourinho et al. 2012). Soil texture and organic matter impact earthworm behaviour and nanoparticle mobility and availability. Clay and organic matter may alter the toxicity of nanoparticles in soil. For nZVI, clay content affects plant toxicity (El Temsah and Joner 2012).

Earthworms help move pore water, air, and solutes to horizontal and vertical soil layers, which increases metal mobility and availability (Leveque et al. 2014). Thus, earthworm bioturbation may facilitate the transport of NMs from bulk soils to the rhizosphere. Rhizosphere microorganisms drive the geochemical behaviour of NMs in the rhizosphere. Earthworms secrete mucus from their body surface, a combination of water, electrolyte, indoleacetic acid, and amino acid (Puga-Freitas and Blouin 2015), which may infiltrate the rhizosphere and interact with NMs. Wang et al. (2020) show that earthworm mucus impacts metal oxide NMs' suspension, hydrodynamic size, and solubility. These active groups of amino acids in mucus (e.g., carboxyl and amino) may speed the dissolution and reduction of NMs, particularly CuO and CeO_2 NMs, and increase their bioavailability (Huang et al. 2017).

The rhizosphere regulates earthworm geochemical behaviour and NM bioavailability in two phases. Physical consequences of earthworm bioturbation on NMs migration from bulk soils to rhizosphere; direct and indirect influences of earthworm mucus on NMs geochemical behaviors in the rhizosphere (Wang et al. 2020). Rhizosphere mechanisms might change NM geochemical processes such as aggregation, anti-aggregation, redox reaction, and transformation, regulating the translocation of NMs into plants and the NMs-mediated physiological and biochemical response of edible crop tissue (Wang et al. 2020). Soil organic matter and root exudates may adsorb on NMs after exposure, changing their hydrophobicity, charge, and metamorphosis (Layet et al. 2017).

A study conducted in Nanjing, China, reveals that TiO_2 and ZnO nanoparticles (NPs) have a detrimental effect on earthworms (Hu et al. 2010). Artificial soil systems containing distilled water, 0.1, 0.5, 1.0, or 5.0 g/kg of NPs were created, and earthworms were exposed for seven days. After an acute toxicity test, Zn and Ti were studied in earthworms, antioxidant enzyme activities, DNA damage, cellulase activity, and gut mitochondria damage. TiO_2 and ZnO NPs may cause substantial harm to earthworms at concentrations over 1.0 g/kg. Ti and Zn, particularly Zn, bioaccumulated, and the most significant dosage in soil (5.0 g/kg) harmed mitochondria. In addition, 5.0 g/kg of ZnO NPs reduced cellulase activity. TiO2 and ZnO NPs are toxic to *E. fetida* at values over 1.0 g kg1 in soil, although ZnO is more toxic (Hu et al. 2010).

Another study in Norway studied nZVI coated with carboxymethyl cellulose on *Eisenia fetida* and *Lumbricus rubellus* utilizing OECD procedures using sandy loam and artificial OECD soil. Earthworms were subjected to 0 to 2000 mg nZVI per kg soil and added fresh or aged in non-saturated soil for 30 days. nZVI concentrations in 500 mg/kg soil negatively influenced earthworm, weight fluctuations, and death. Furthermore, 100 mg/kg nZVI also influenced reproduction. nZVI's toxicity was decreased after some time, with significant variations between old and non-aged soils (El-Temsah and Joner 2012).

No meta-analysis studies addressed NMs-mediated changes in the rhizosphere soil environment (Wang et al. 2020). Furthermore, the control of soil fauna and earthworms on the chemical behaviour of NMs in the rhizosphere is still in its infancy; hence more advanced studies for better integration of earthworms into the geochemical behaviour of NMs driven by rhizosphere processes are needed in the future (Wang et al. 2020).

8. Existing and possible effects of nano-formulated agrochemicals on millets

Nanomaterials are used as nano fertilizers, nano pesticides, and nanosensors to boost agricultural output. Engineered nanomaterials (ENMs) affect fertilizer nutrient availability in soil and plant uptake. These materials can control crop diseases by directly acting on pathogens and generating reactive oxygen species (ROS). ENMs may potentially reduce disease by boosting crop nutrition and plant defense. In addition, ENMs may supplement traditional fertilizers and insecticides, lowering agricultural and environmental impact. Usually, millets utilize significantly less quantity of fertilizers and other agrochemicals. In India, it is mainly grown in the rainfed areas during the summer season; habitually, the field and agronomic operations are much less than other food crops (Faiz et al. 2022). Here are 10 Millets that are eaten globally: The major millets 1. Finger Millet (Ragi), 2. Foxtail Millet (Kakum/Kangni), 3. Sorghum Millet (Jowar), 4. Pearl Millet (Bajra), 5. Buckwheat Millet (Kuttu), 6. Amaranth Millet (Rajgira/Ramdana/Chola), 7. Little Millet (Moraiyo/Kutki/Shavan/Sama), 8. Barnyard Millet, 9. Broomcorn Millet and 10. Kodo Millet (Srivastava and Bisht 2021)

8.1. Effect of nano formulation on disease suppression on millets

Chitosan nanomaterials (CNM) delayed blast disease signs in infected finger millet from 15 to 25 days and decreased the severity by 64% on day 50. Increases in peroxidase activity and ROS generation led to disease suppression (Sathiyabama and Manikandan 2016). This shows that CNMs may boost plant growth during disease attacks via generating ROS. In other plant-pathogen, downy mildew disease induced by *Sclerospora graminicola* was significantly decreased (82%) in millet seeds exposed to CNMs (250 mg/kg soil), although less effective than the commercial fungicide (metalaxyl was 92% effective). 143 Ammonia lyase, catalase, peroxidase, polyphenol oxidase, phenylalanine, and superoxide dismutase increased in treated plants, suppressing disease severity (Siddaiah et al. 2018). Like finger millet, a Cu–chitosan nanocomposite boosted maize's defenses against *Curvularia lunata* (Sathiyabama and Charles 2015) and *Fusarium verticillioides* (Adisa et al. 2019). Some studies reported that nanocomposite improved agricultural yields (Choudhary et al., 2017). Combining seed treatment with foliar nanocomposite was more effective than foliar treatment alone against *P. grisea* in finger millet. The seed treatment was proposed to safeguard the plant by boosting defense enzymes (Adisa et al., 2019).

Pre-soaked seeds in chitosan nanoformulation at the rate of 0.1% w/v in Finger millet suppresses the growth and incidence of *Pyricularia grisea* and blast disease by 64% at day 50 after inoculation (Sathiyabama and Manikandan 2016). Nano formulation of Selenium with *Trichoderma asperellum* at the rate of 150–250 mg/L on pearl millet in the form of Foliar spray suppresses the downy mildew disease *Sclerospora graminicola.* In addition, it improves early plant growth (Nandini et al., 2017). Pre-soaked seed treatment and foliar treatment at the rate of 0.1% w/v concentration of Copper (II) oxide (CuO) on Finger millet has a positive effect on controlling the disease in *Pyricularia grisea* and blast disease incidence by 75%. As CuO can also act as a copper micronutrient, it increased the number of leaves by 22% when used as a foliar spray and by 33% as combined exposure of seed soaking and foliar spray. It also increased the defense enzymes in the finger millet (Sathiyabama and Manikandan 2018).

8.2. Effect of nanoformulations as nutrition and fertilizer on millets

Application of Zn-nano fertilizers on six-week-old finger millet plants improved shoot length (15.1%), root length (4.2%), root area (24.2%), chlorophyll content (24.4%), total soluble leaf protein (38.7%), dry plant biomass (12.5%), and enzyme activities of acid phosphatase (76.9%), alkaline phosphatase (61.7%), phytase (322.2%), and dehydrogenase (21%). In addition, zinc nano fertilizer increased crop output by 37.7% (Tarafdar et al., 2014).

Plant height, thousand-grain weight, and grain yield quantitative characteristics of foxtail millet did not change statistically between ZnO NP-treated and untreated plants; however, ZnO NP-treated plant grains had considerably greater oil and total nitrogen contents and reduced crop water stress index (CWSI) (Kolenčík et al. 2019). The slow-releasing nano-fertilizer increases plant physiological qualities and grain nutritional parameters, which benefits nanomaterial-based companies. In addition, the photocatalytic impact of ZnO NPs on leaf surfaces speeds up photosynthesis (Kolenčík et al., 2019; Morab et al., 2021).

In a study conducted in Haryana, nitrogen and zinc nano fertilizers were employed with organic farming on wheat, pearl millet, mustard, and sesame. Organic farming using nitrogen and zinc nano fertilizers increased wheat, sesame, pearl millet, and mustard yields by 5.35, 24.24, 4.2%, and 8.4%, respectively (Kumar et al. 2022). Increased yield correlated with wheat tillers, pearl millet ear head length, sesame capsule number per plant, and mustard siliquae number per plant. Organic manure, bio-fertilizer, and nano fertilizers produced greater yields and better plant development than standard chemical fertilizers alone. The findings imply that nanotechnology and organic farming may reduce the need for chemical fertilizer while increasing crop yield.

8.3. Nanoformulations as pesticides on millets

Nanotechnology is a potential crop protection technique, especially in lowering the quantity of active ingredients (AI) used in the field. Despite this technology's safety, efficiency, sustainability, and field performance, a big regulatory concern persists. Despite the promises and problems of developing, scaling up, and applying nanoformulations in the field, the worldwide market indicates a consistent expansion of this technology and a push toward innovation by employing sustainable components and new application models (Ferreira et al. 2022). Nanopesticides have the potential to be safer and more effective than their traditional pesticides. However, nano pesticides affect insect physiology and metabolism in plants during their interactions with them, especially regarding their method of action, which is little known. The current primary concern of scientists is to study the interaction of nano pesticides with plants.

Inorganic, porous nanoparticles may help deploy nanotechnology in the form of nano pesticides. Inorganic porous nanoparticles may encapsulate and transport insecticides and fertilizers. Porous nanocarriers may boost pesticide solubility and mobility and manage their release. Silica, iron oxide, zinc oxide, copper oxide, clays, and hydroxyapatites are common inorganic nanomaterials researched for crop development and protection (Bueno and Ghoshal 2022). This approach may give shelf-stable formulations, improved field performance with a lower AI dosage, and fewer tank combination difficulties. Nanoformulations applied by drones are a developing technology in precision agriculture that may expand their usage (Ferreira et al., 2022).

Copper-based nanoparticles offer minimal manufacturing costs and powerful antibacterial activities at acceptable doses, making them excellent for agricultural applications. Copper-based nanomaterials may be made by classic chemical, physical, and biogenic processes. Copper aids plant development, metabolism, and defense, and it's used in fungicides to fight plant diseases. In addition, copper-based nanoparticles have created new ways to protect and defend crops, with better outcomes and lesser toxicity than conventional copper (Gomes et al. 2022).

A nanosystem comprising chlorpyrifos (CPF), polydopamine (PA), attapulgite, and calcium alginate (CA) was used to create pH-responsive controlled release chlorpyrifos (PRCRC). The mechanism protects CPF molecules from UV radiation. PA-CA hydrogel showed good biosafety on *Escherichia coli* and foxtail millet. It is a promising strategy to increase pesticide consumption efficiency and longevity, which has an extensive application opportunity in the future (Xiang et al. 2018).

Silver-doped MoO3 and h-MoO3 nanoparticle hydrothermal production, characterization, and biological applications (NPs). X-ray diffraction and vibrational spectrum measurements revealed the NPs' phase forms. *Staphylococcus aureus*, *Bacillus cereus*, *Citrobacter koseri*, and *Pseudomonas aeruginosa* were used to investigate the antibacterial activity of the nanoparticles. In addition, foxtail and finger millet seed germination was tested over seven days and found the antimicrobial properties (Raj et al., 2021).

8.4. Nanoformulation for toxin reduction in millets

Montmorillonite clay's availability and environmental friendliness make it helpful in adsorbing mycotoxins in food and feed. For example, Zearalenone (ZEA) is an oestrogenic mycotoxin linked to hormonal and reproductive problems in humans and animals. Montmorillonite clay and lemongrass extract nanoformulations were produced to reduce millet toxicity. It was applied to 8 and 12 percent millet samples and kept for four weeks. XRD patterns for Montmorillonite clay compositions showed Montmorillonite peaks at 20.06° and quartz at 26.56° and 68.53°. After four weeks, ZEA levels in all treated samples were measured using LC-MS/MS. All compositions removed zearalenone. After four weeks, *Cymbopogon citratus* powder decontaminated zearalenone in cereal (Olopade et al., 2019).

8.5. Nanoformulation for drought tolerance in millets

Underwater deficiency conditions, chitosan nano-emulsion was tested on pearl millet in Tamil Nadu, India. At the blooming stage, under moisture stress, 0.1 percent chitosan nanoemulsion was foliar sprayed. The treatment with chitosan reduces stomatal conductance, photosynthesis, transpiration, and leaf temperature. However, under drought, treated plants produced more than controls. The photosynthetic rate, stomatal conductance, and transpiration rate were all reduced by 4.89 percent, 6.75 percent, and 7.97 percent, respectively, on the first day following foliar application chitosan under water deficit conditions. Hence, the foliar spray of chitosan alleviates water stress by enhancing plant water status and productivity (Priyaadharshini et al., 2019).

8.6. Nanoformulation from millets as human protein and foods

Using ultrasonication, crocin-loaded (Saffron active) protein nanoparticles were produced from sorghum (SPCN), foxtail millet (FPCN), and pearl millet (PPCN). Nano-sized crocin-protein complexes with decreased polydispersity index and negative zeta potential were created. Crocin encapsulation efficiency was 83.78 (FPCN), 78.74 (SPCN), and 70.01 (NPCN) (PPCN). FE-SEM pictures of crocin-loaded protein nano complex topography (FE-SEM). ATR-FTIR study confirmed nanoencapsulation by showing crocin peaks at 956, 1700, and 3350 cm^{-1} in protein-crocin nanocomplex. The antimicrobial activity of crocin-loaded protein nano complex against *E. coli, S. aureus, and F. oxysporium* was also investigated. Protein nanoparticles may operate as an effective wall material since they maintain crocin's bioactivity (anti-cancerous and anti-hypertensive) after in-vitro digestion. Protein nanoparticles demonstrated high encapsulation effectiveness for crocin utilizing ultrasonication and performed as an efficient carrier/wall material for bioactive, keeping them from degradation during gastrointestinal transit and preserving bioactivity. The nanocomplex may be used to strengthen future functional foods (Jhan et al., 2022).

8.7. Use of nano fertilizers in agriculture for long-term sustainability

Nutrient fertilization improves crop yield and quality by preserving soil fertility. Chemical fertilizers make precise nutrient management of horticulture crops a global concern. However, traditional fertilizers may be detrimental to persons and the environment. Therefore, nanotechnology is developing as a possible option for ecologically friendly fertilizers with excellent nutrient-use efficiency (Zulfiqar et al., 2019).

Nano fertilizers may boost nutrient usage efficiency, benefiting nutrition management. Nutrients attached to nano-dimensional adsorbents release nutrients more slowly than traditional fertilizers. This strategy increases nutrient-use efficiency and reduces groundwater leaching. Nano fertilizers may also improve abiotic stress tolerance, and when combined with microorganisms (nanobiofertilizers), they give additional advantages (Zulfiqar et al. 2019).

Nano fertilizers provide new sustainable agriculture methods; however, their limits should be recognized before commercial deployment. Nanomaterials released into the environment and food chain may harm human health. Nano fertilizers provide enormous prospects to increase plant nutrition and stress tolerance to produce greater yields in a climate change context; however, not all nanomaterials are safe for all uses. Nano fertilizer dangers should be thoroughly evaluated before usage, and biotechnology improvements are needed for safe use in agriculture (Zulfiqar et al., 2019).

9. Advantages and disadvantages

There is enormous pressure on the agriculture sector to find a solution for the continuous and bulky usage of chemicals in agriculture productivity to feed the growing population. This is because the nutrients applied to the soil are leached or lost than absorbed by plants. On the other hand, the soil rhizosphere communities of millets and other crops require fertilizers to retain their metabolisms and growth while increasing crop productivity. The development of nanoformulations with crucial properties is gaining attention worldwide; therefore, both sides of the coin need to be addressed for sustainable usage **(Fig. 4).**

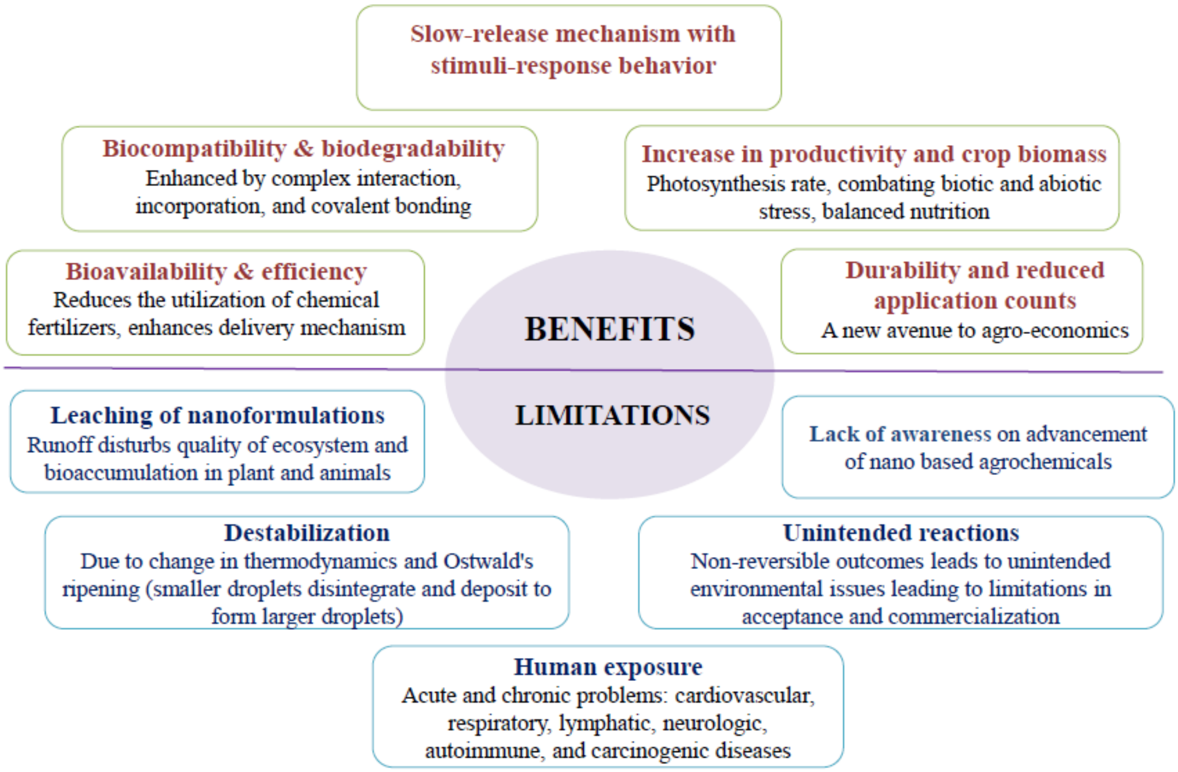

Fig.4. Benefits and limitations of nano-based agroformulations

10. Research gaps and future perspective

In agriculture, more than 232 nano-enabled products are manufactured by 75 companies in 26 countries worldwide, most of which are in the fertilizer and pesticides category (Vishnu and Dhandapani 2021). The optimal NPK fertilizers ratio of 4:2:1 is ideal for crop productivity, while the current ratio is maintained at 10:2.7:1 in India, with more heavily subsidized urea (Sharma et al. 2021). A clear understanding of the interaction between nanoparticles and rhizosphere reactions is necessary to assess the impact of nano-formulated agrochemicals on crop production through improved disease resistance, enhanced nutrient use efficiency, and better crop yield, mainly in millets. The interaction of nano-formulated agrochemicals in millets crops should have a better understanding of regarding the important aspects such as (i) mechanism behind the improvement of growth, resistance, and yield of millets, (ii) impact on the rhizosphere environment, (iii) adverse effects on millets if any. However, extensive research in millet crops regarding the impact of nano-formulated agrochemicals is needed. It is also essential to consider the factors such as environmental safety, cost efficiency, sustainability, and availability to farmers for large-scale applications.

As a result, fresh ways are required to safeguard the food plants with minimum or no adverse environmental consequences. Developing several nano-formulated agrochemicals has demonstrated higher agro-chemical activity, controlled release capacity, and targeted delivery qualities. They can also restore the oxidative stress symptoms induced by numerous stresses in roots and plants and stimulate plant adaptation to unfavorable conditions, making them an important agricultural tool. Though nanotechnology finds a vast relevance in agriculture, it is still in a very nascent stage, especially on nano biomaterials for agriculture use. The nano products developed so far have not yet penetrated the market and have not reached the common farmer. Concerted efforts are needed to promote the application of nanotechnology with multi-stakeholder participation, including the Government, Private Sector industries, educational institutions, and other relevant stakeholders engaged in agriculture. Formulating suitable policies and educating and encouraging the farmers for wide-scale use will sustainably assist in food and nutritional security, contribute to a safeguarding environment, and combat climate change.

11. Conclusion

The chapter discussed the state of nanoformulations in rhizospheric communities focusing on the current status and possible effects on rhizosphere communities of millets. This groundwork on millets will help formulate further research on the influence of nanoformulations of microbial communities in millet farming and how it may pose challenges to humans through the food chain. The alteration in soil microbiota such as microbial biomass, the microbial population including rhizosphere communities, enzyme activities, and function are positively explored during nano agrochemical application. But the uncertainty of nanomaterials in agriculture must be explored more, and a life cycle assessment of each nano product should be discovered before commercialization. The possible effects of millet rhizosphere communities may differ based on the dosage and concentration of nanoformulations and synthesis method. Hence, exploration of toxicity on the environment should be a research priority, especially in millets, since research findings on millet rhizosphere communities are meager comparatively. Biosynthesis of nano agrochemicals should be encouraged to improve yield and productivity sustainably.

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